

AASTHA SHAH

Pune, Maharashtra

☎ +91-9511886941

✉ aasthashah21715@gmail.com

🌐 LinkedIn

🐙 GitHub



EDUCATION

Pune Institute Of Computer Technology — BE 9.22

2022–2026

TECHNICAL SKILLS

Technical Skills: C, C++, Java, Python, JavaScript, Angular, Flask, FastAPI, HTML, CSS, SQL, NoSQL, DSA, Git/Github

Soft Skills: Problem Solving, Communication, Time Management, Attention to Detail, Critical Thinking, Leadership, Adaptability, Agile Collaboration

COURSEWORK

- Data Structures & Algorithms
- Operating Systems
- Computer Networks and Security
- Database Management Systems
- Object-Oriented Programming (OOPS)
- Web Technology
- Artificial Intelligence
- Machine Learning

INTERNSHIP

BNY — SDE Intern

Pune, India

- Secured 3rd place in an internal Fixathon, focusing on identifying and remediating IDOR, component-level, and XSS vulnerabilities using tools such as Veracode, VMAD, and sandbox scans.
- Enhanced a Python utility to automate template updates, achieving a 95% success rate by programmatically replacing old company logos with new ones across 30K+ of DOC templates with varying formats.
- Developed to production-grade Python code, emphasizing secure and scalable development practices, while proactively learning backend and DevOps tools like Spring Boot, Docker, and CI/CD pipelines.
- Collaborated in an agile team environment, showcasing strong debugging skills and secure coding standards using GitLab workflows

Enthral.ai — SDE Intern

Pune, India

- Built responsive frontend components using React and AngularJS, improving UX and performance for a live proctoring platform. Developed a WebSocket-based proctoring system using FastAPI for backend.
- Worked extensively on real-time video streaming, API integration, authentication mechanisms, and multi-camera synchronization for remote proctoring.
- Integrated machine learning models into the system pipeline for anomaly detection during exams.
- Implemented Celery with Redis for asynchronous task handling, significantly improving concurrency, responsiveness, and reliability in real-time data processing.

PROJECTS

Vulnerability Management and Assessment System | GitHub Actions, Docker, Kubernetes

Present

- Developing an open-source-driven cybersecurity platform integrating DevSecOps for full lifecycle vulnerability management — from detection and validation to autonomous remediation and compliance verification.
- Using GitHub Actions for CI/CD automation which enables organization-level reusable workflows for consistent and scalable security scanning across repositories.
- Integrated multi-domain vulnerability scanners (SonarQube, Trivy, Bandit, etc) within GitHub workflows to ensure continuous and automated code analysis.
- Implementing AI-powered remediation agents that analyze scan reports and generate secure patches automatically.
- Designing secure report pipelines to store scan results in AWS S3 and compliance mappings in PostgreSQL, with role-based validation and governance workflows.

Secure Data Wiping Tool for Recycling of IT Assets | C

Present

- Designing and developing a secure, cross-platform data wiping application to enable verifiable, tamper-proof erasure of sensitive data from HDDs, SSDs, and hidden storage areas.

- Built on ShredOS/Linux framework with a custom GTK GUI replacing command-line tools for one-click usability.
- Implementing hardware-based erase commands with fallbacks to multi-pass overwriting for full NIST SP 800-88 compliance.
- Integrating post-erasure verification tool to automatically confirm irrecoverable data deletion.
- Automating generation of digitally signed wipe certificates (PDF and JSON) for auditable proof.
- Optimized for offline usage (bootable USB/ISO) and third-party verification of wipe status.

Database System Using B+ Tree with Concurrent Retrieval | Java 2024

- Designed and implemented a high-performance database system using the B+ Tree data structure to efficiently manage and store student records.
- Focused on fast and concurrent data retrieval using Java multithreading with locks and latches to handle a high volume of read/write operations.
- Utilized the properties of the B+ Tree to support quick search operations while maintaining balanced tree structure and efficient indexing.
- Implemented persistent storage using object serialization to ensure data integrity and consistency across sessions.

RESEARCH

Extractive Text Summarization using Extended TextRank Algorithm 2024

- Conducted research to improve ROUGE scores in extractive text summarization using NLP techniques under the guidance of Dr. Sheetal Sonawane.
- Enhanced the base TextRank algorithm by introducing a dynamic factor to increase entropy and capture topic diversity.
- Published work at ICON 2024; paper available at: aclanthology.org/2024.icon-1.54.pdf